PlantTraitNet: An Uncertainty-Aware Multimodal Framework for Global-Scale Plant Trait Inference from Citizen Science Data

Ayushi Sharma, Johanna Trost, Daniel Lusk, Johannes Dollinger, Julian Schrader, Christian Rossi, Javier Lopatin, Etienne Laliberté, Simon Haberstroh, Jana Eichel, Daniel Mederer, Jose Miguel Cerda-Paredes, Shyam S. Phartyal, Lisa-Maricia Schwarz, Anja Linstädter, Maria Conceição Caldeira, Teja Kattenborn

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Motivation

Data

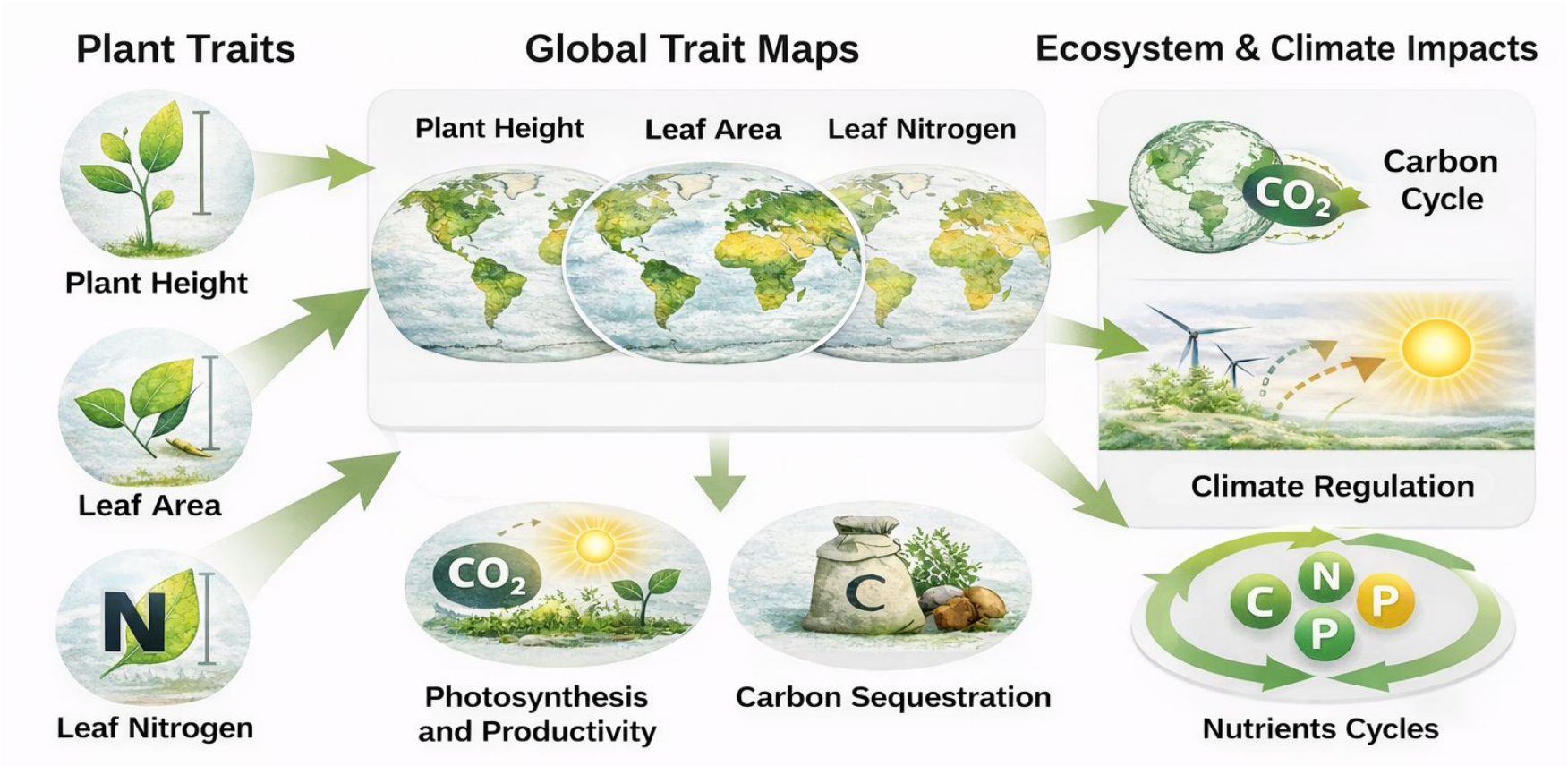
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Plant Traits
Global Trait Maps
Ecosystem & Climate Impacts
Plant Height
Leaf Area
Leaf Nitrogen
Plant Height
Leaf Area
Leaf Nitrogen
CO2
Carbon Cycle
Climate Regulation
CO2
C
C
N
P
P
Photosynthesis and Productivity
Carbon Sequestration
Nutrients Cycles

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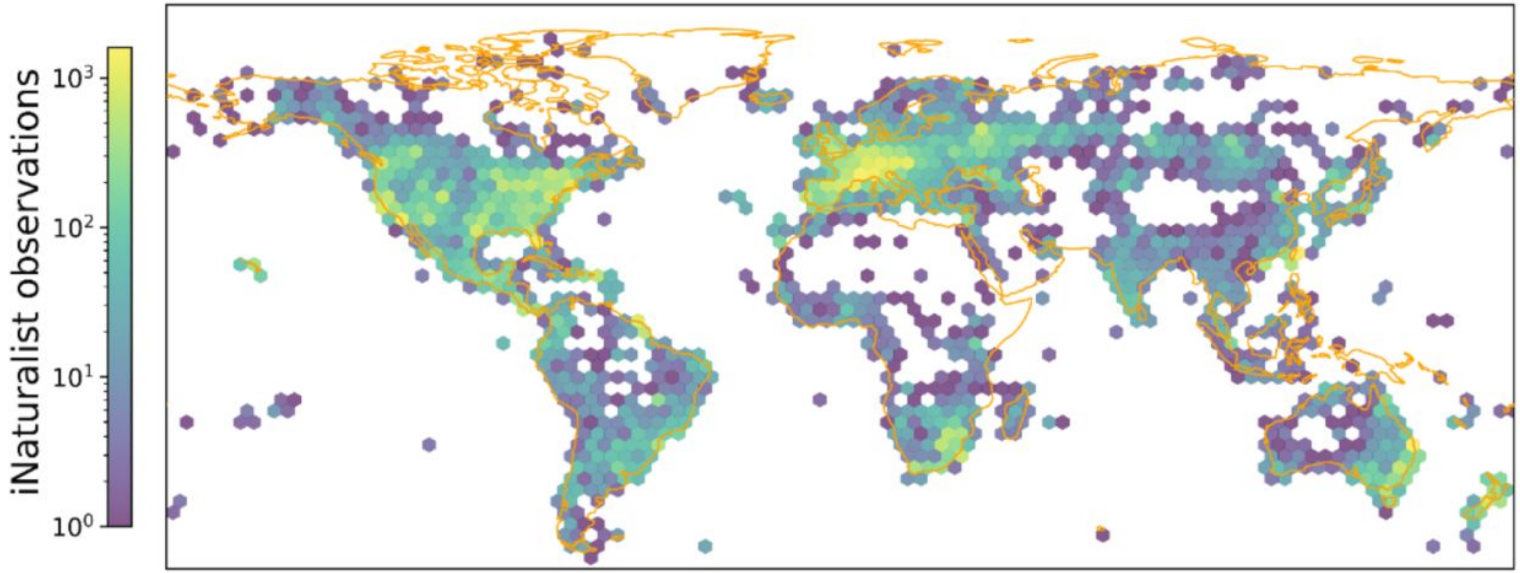
Data Based on Citizen Science Contributions

iNaturalist Dataset: Georeferenced Plant Images with Species Labels



Data Based on Citizen Science Contributions

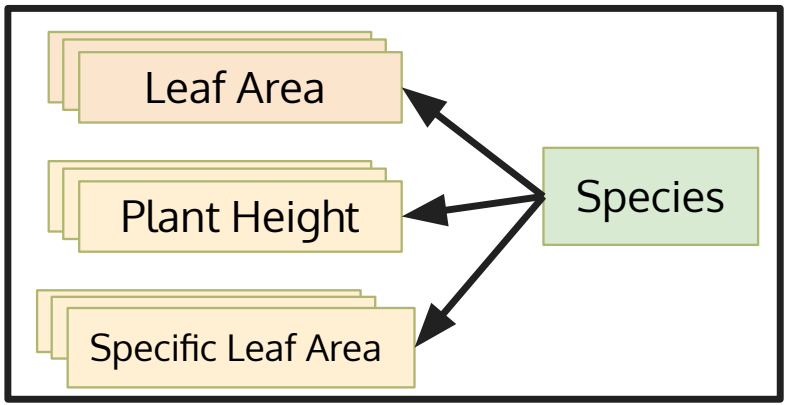
iNaturalist Dataset: Georeferenced Plant Images with Species Labels



Reference: [GBIF - iNaturalist Research-grade Observations]

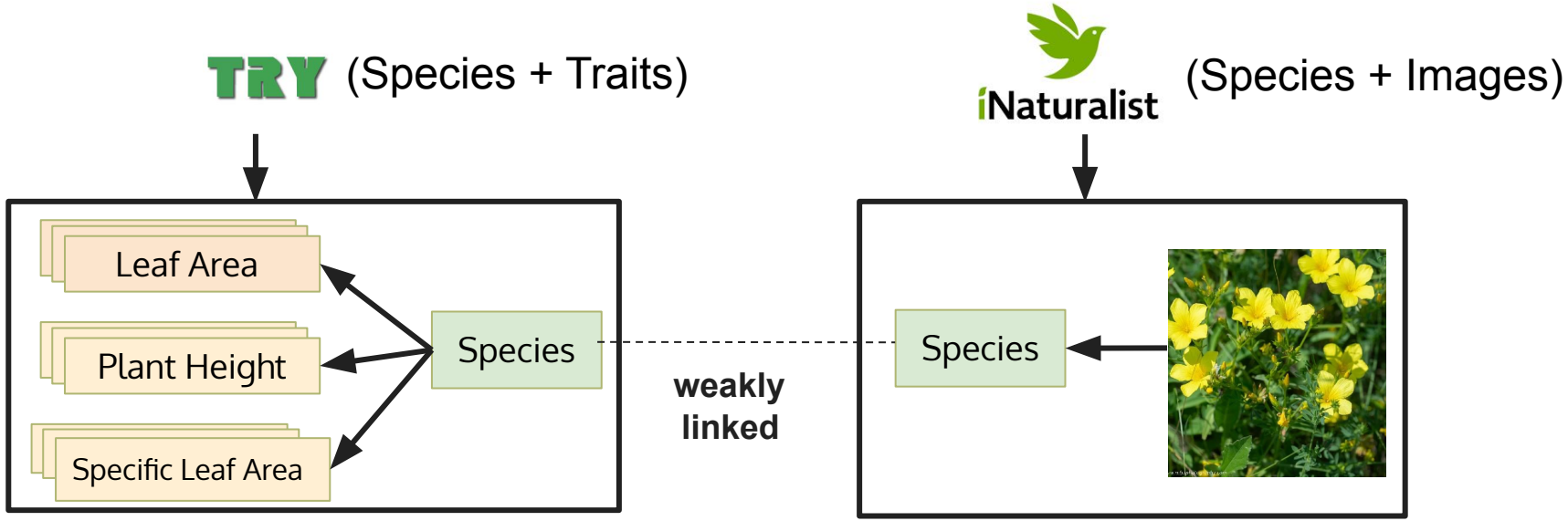
TRY Database: Plant Trait Data

TRY Database: Species Occurrences and Plant Trait Measurements

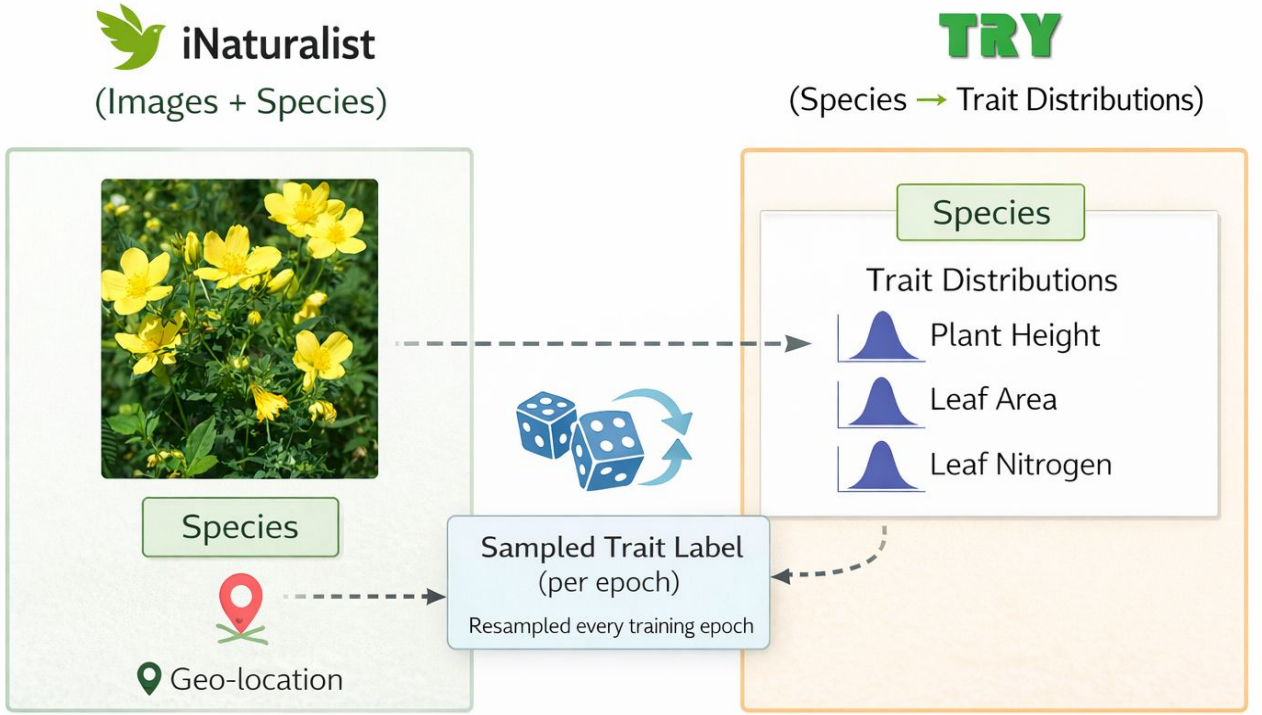


Reference: TRY—A plant trait database of databases

Weak Annotation



Weak Annotation

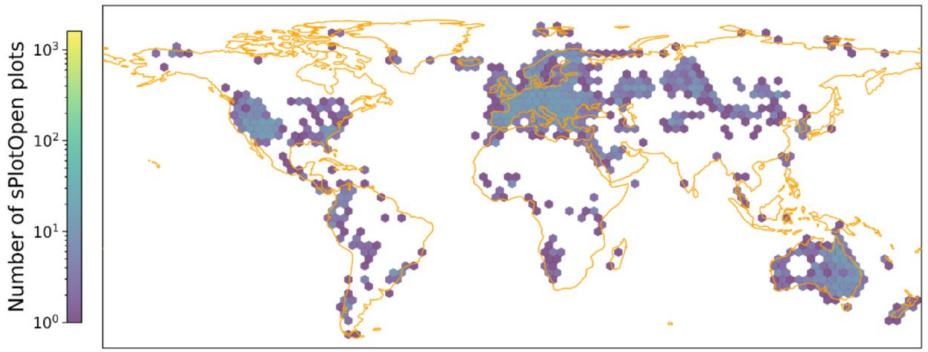


Weak Trait Annotation of Plant Images via Species-Level Trait Distributions

sPlotOpen: A Global Dataset of Vegetation Plots

Obtain data on species communities across space

- Compute community-weighted trait values using TRY
- Sparse spatial coverage
- Reliability of derived trait estimates



Reference: [sPlotOpen – An environmentally balanced, open-access, global dataset of vegetation plots]

Overview of Data

Training Data

- iNaturalist Dataset: Georeferenced Plant Images with Species Labels
- TRY Database: Plant Trait Data

Reference Data

- sPlotOpen (Community weighted trait values)

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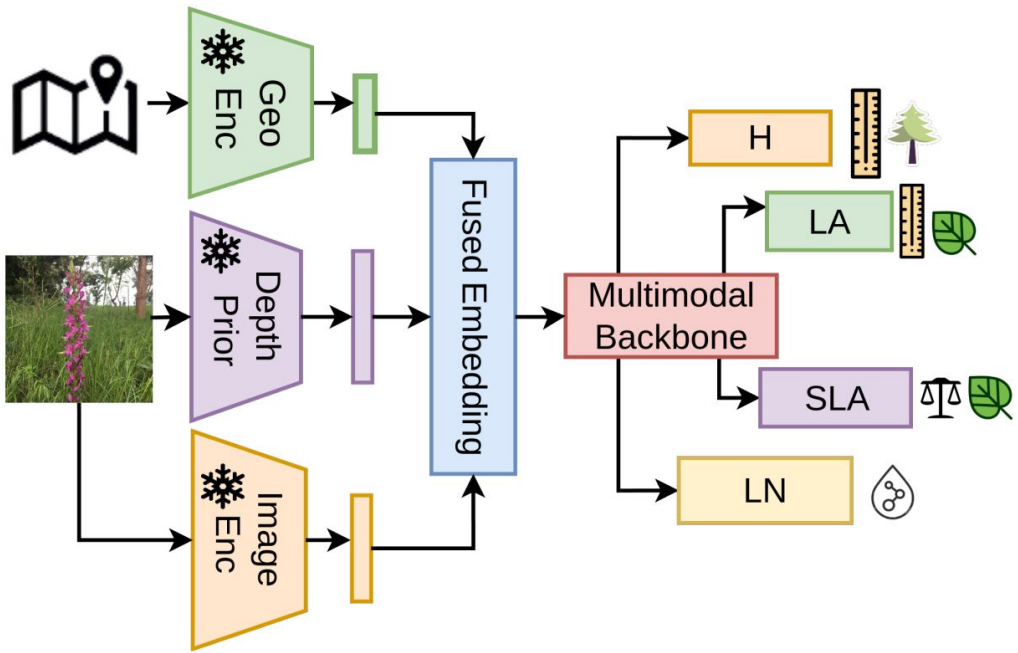
Conclusion

Model

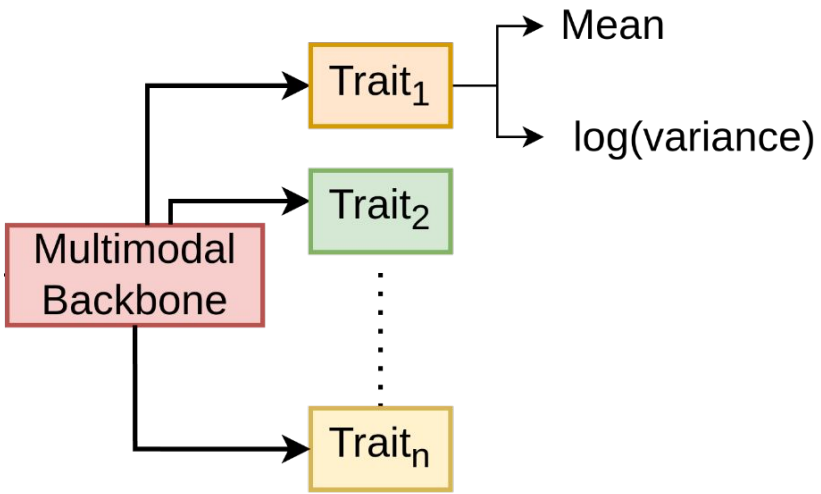
Image
- Image Encoder: Vit-b from DINOv2
- Depth Encoder: Depth Anything V2 - Vit-b

GeoLocation
- Location encoder from Climplicit model

Multimodal Backbone
- 8 Resnet blocks



Model Training



High variance, High uncertainty

Negative Log-Likelihood Loss (NLL) :

$$\mathcal{L}_{\mathrm{NLL}}^{m} = \frac{1}{N}\sum_{n=1}^{N}\left[\frac{(y_n^m - \hat{\mu}_n^m)^2}{2\exp(\hat{s}_n^m)} + \frac{1}{2}\hat{s}_n^m\right]$$

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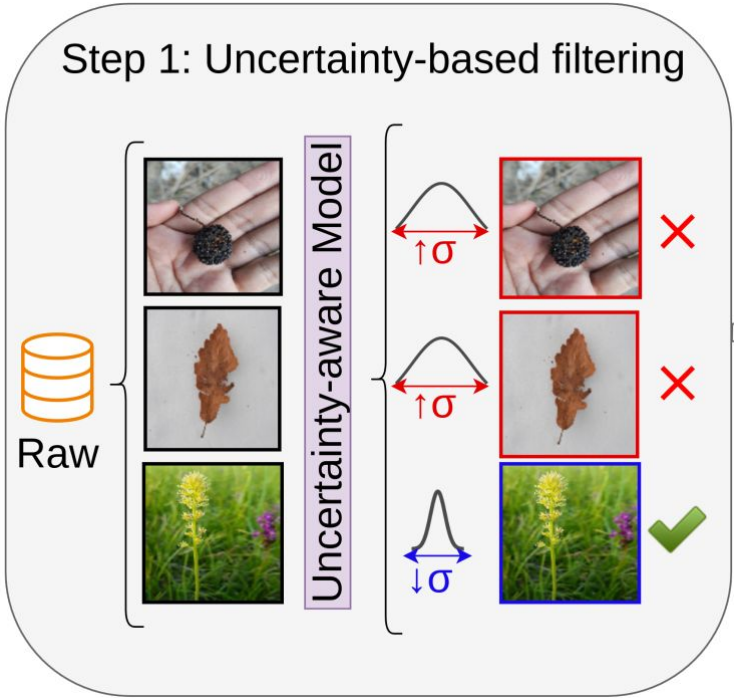
Challenge 1: Noisy Plant Images

- **Plant Height**: High uncertainty from unsuitable visual contexts, such as winter scenes, presence of fruits, or human hands
- **Specific Leaf Area (SLA)**: Uncertainty for images with bare branches, flowers, or buds
- **Leaf Nitrogen**: High uncertainty for low-quality or blurry images



Step 1: Uncertainty-aware Filtering

- Images with high predictive uncertainty are excluded to improve label reliability.



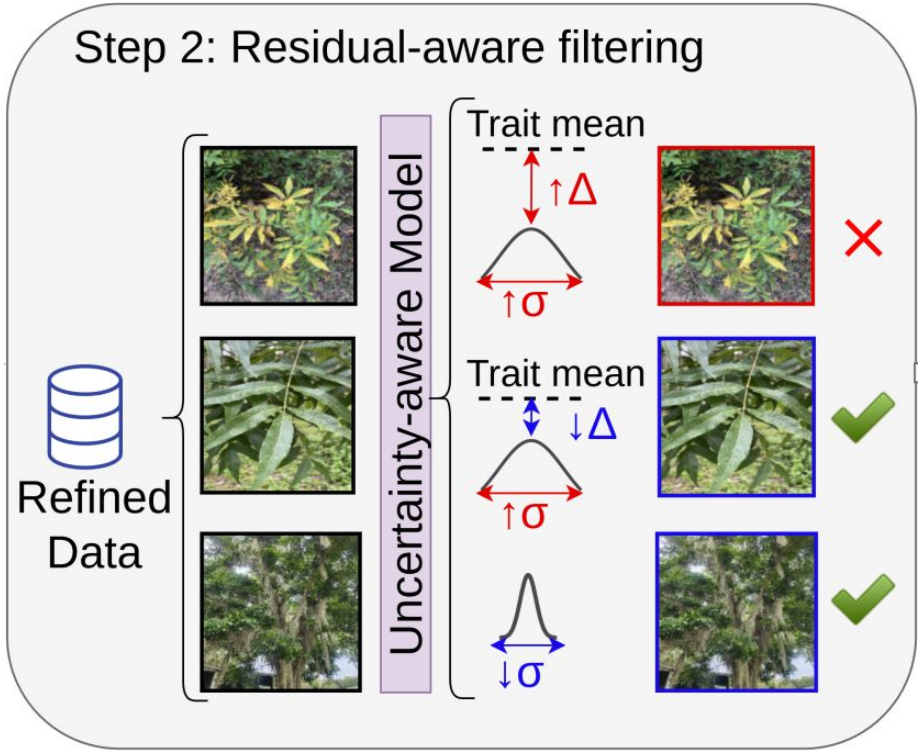
Limitation - Heteroscedasticity



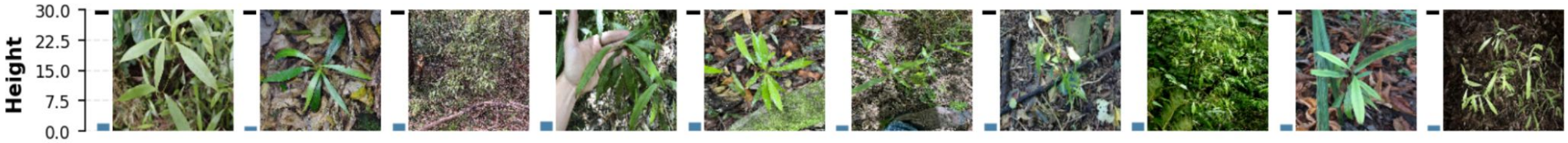
Higher uncertainty for taller plants

Step 2: Residual-aware Filtering

- Remove images with high uncertainty and high residual error
- Helps to mitigates extreme noisy labels

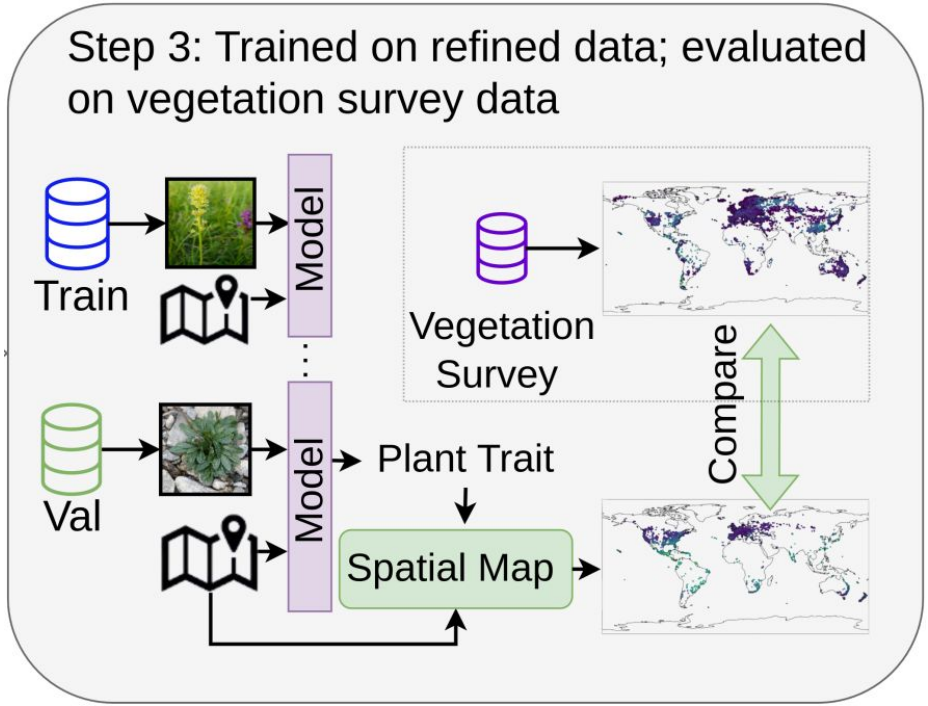


Residual-aware Filtering

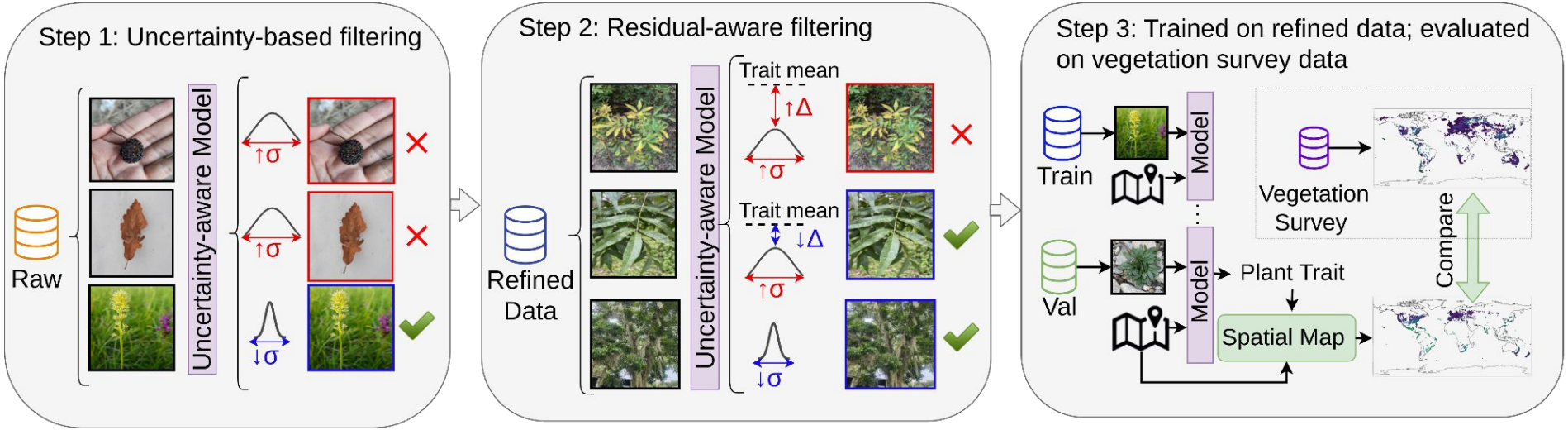


Step 3: Training and evaluation

- Train the model
- Aggregate spatially using geolocation
- Evaluate against sPlotOpen



Overview: An Uncertainty-aware Framework



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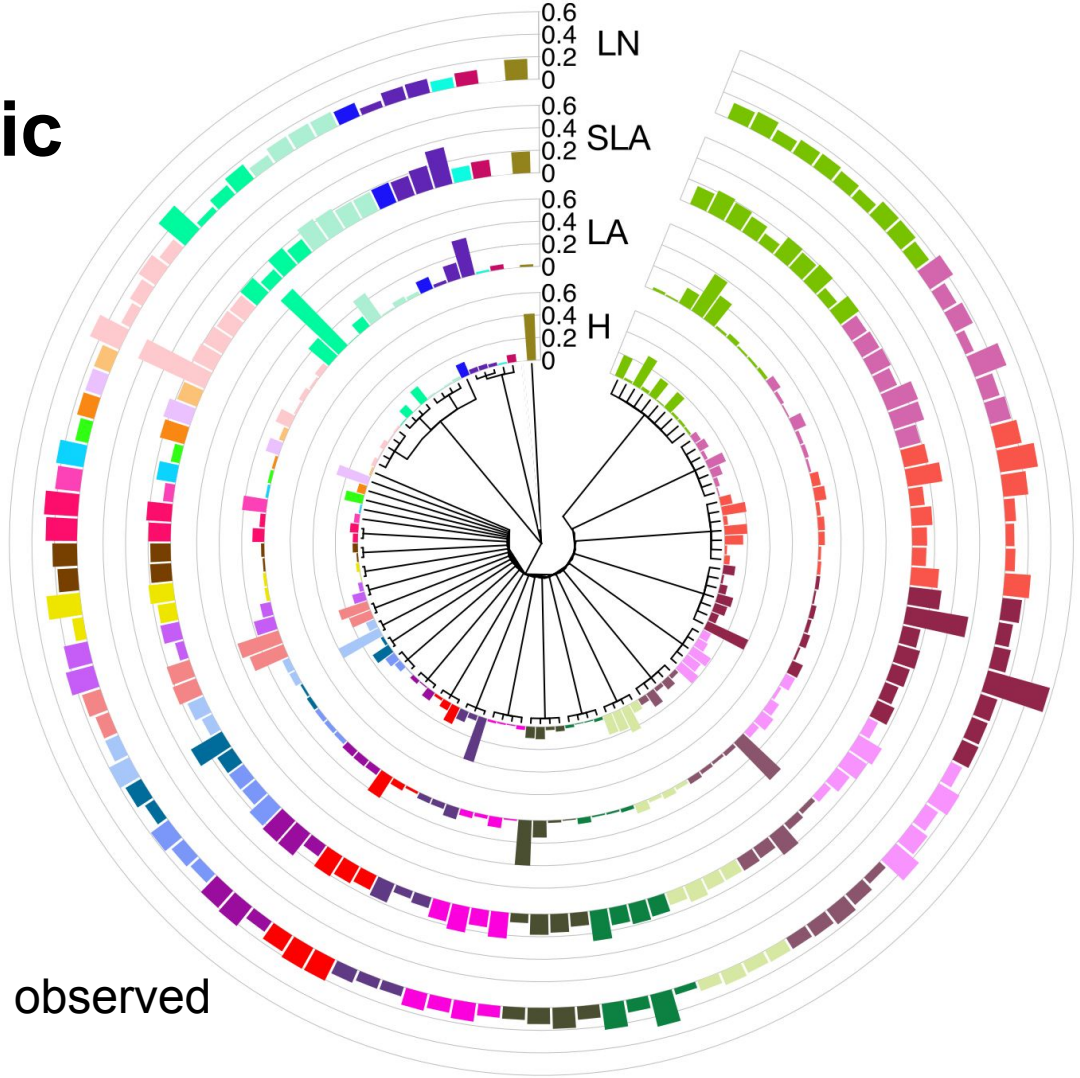
Baseline

Takeaway:

- Uncertainty-aware filtering helps in performance improvement for most traits
- PlantTraitNet outperforms previous methods for plant height, leaf area and specific leaf area

Method	Metric	H	LA	SLA	LN
PlantTraitNet (Raw)	$R^2 \uparrow$	**0.19**	*0.30*	*0.23*	-0.16
	nMAE $\downarrow$	**0.22**	**0.14**	*0.14*	*0.17*
	$r \uparrow$	**0.45**	*0.56*	**0.59**	*0.49*
PlantTraitNet (Refined)	$R^2 \uparrow$	*0.18*	**0.34**	**0.27**	*-0.12*
	nMAE $\downarrow$	**0.22**	**0.14**	**0.13**	*0.17*
	$r \uparrow$	**0.45**	**0.57**	**0.59**	**0.50**
Schiller et al., 2021	$R^2 \uparrow$	-0.32	0.11	0.16	**0.06**
	nMAE $\downarrow$	*0.28*	*0.17*	*0.14*	**0.14**
	$r \uparrow$	0.42	0.52	*0.53*	0.40
Wolf et al., 2022	$R^2 \uparrow$	-0.61	-0.02	0.02	-0.20
	nMAE $\downarrow$	0.31	0.18	0.16	0.18
	$r \uparrow$	*0.43*	0.53	0.50	0.41
Madani et al., 2018	$R^2 \uparrow$	–	–	-0.72	-0.85
	nMAE $\downarrow$	–	–	0.23	0.22
	$r \uparrow$	–	–	0.23	0.17
Butler et al., 2017	$R^2 \uparrow$	–	–	-0.17	-0.50
	nMAE $\downarrow$	–	–	0.18	0.20
	$r \uparrow$	–	–	0.29	0.32
Boonman et al., 2020	$R^2 \uparrow$	–	–	0.03	-0.37
	nMAE $\downarrow$	–	–	0.16	0.18
	$r \uparrow$	–	–	0.49	0.20
Madani et al., 2018	$R^2 \uparrow$	–	–	-0.76	–
	nMAE $\downarrow$	–	–	0.23	–
	$r \uparrow$	–	–	-0.07	–
van Bodegom et al., 2014	$R^2 \uparrow$	–	–	-1.00	–
	nMAE $\downarrow$	–	–	0.24	–
	$r \uparrow$	–	–	0.33	–

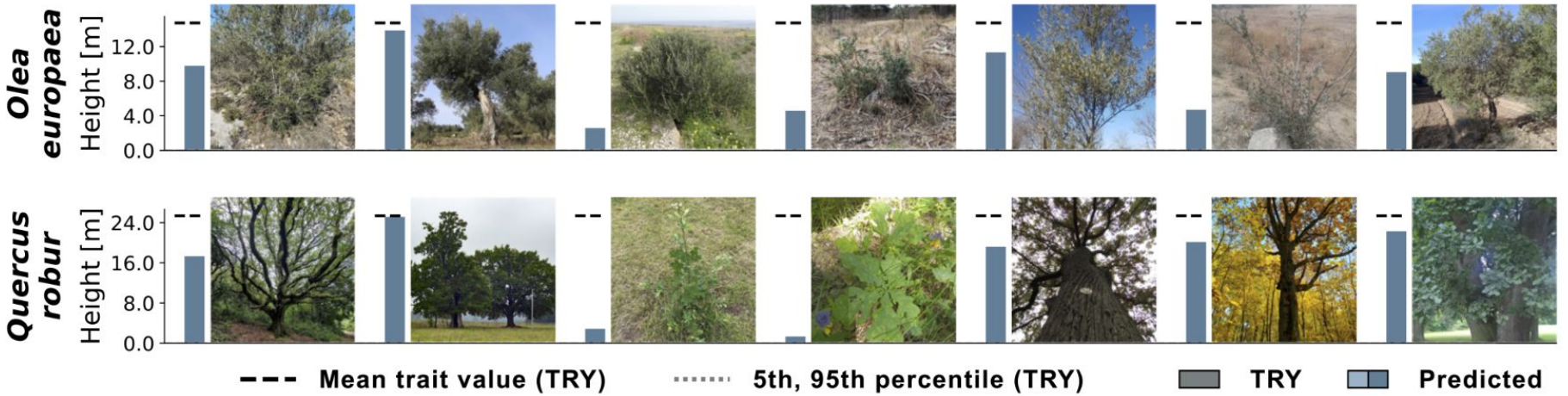
Phylogenetic Correlation





Takeaway:
No systematic bias observed

Intra-species Variation



Takeaway:
Model predictions capture trait variability within the species.

Ablation

	Image	Geo	Depth	H	LA	SLA	LN	# Top ranks
Multi-Task	DinoV2	✗	✗	0.15 ± 0.00	0.31 ± 0.00	**0.32 ± 0.00**	0.14 ± 0.01	1
	BioCLIP	✗	✗	0.15 ± 0.01	0.3 ± 0.00	**0.32 ± 0.01**	0.15 ± 0.04	1
	DinoV2	SatCLIP	✗	0.16 ± 0.02	0.27 ± 0.04	0.25 ± 0.02	0.11 ± 0.05	0
	DinoV2	GeoCLIP	✗	*0.17 ± 0.01*	**0.33 ± 0.01**	**0.32 ± 0.00**	0.15 ± 0.02	3
	DinoV2	Climplicit	✗	**0.19 ± 0.01**	*0.32 ± 0.01*	*0.31 ± 0.01*	0.16 ± 0.06	3
	BioCLIP	Climplicit	✗	**0.19 ± 0.00**	*0.32 ± 0.02*	*0.31 ± 0.01*	0.15 ± 0.06	3
	BioCLIP	Climplicit	DA-V2	0.16 ± 0.01	0.28 ± 0.03	0.30 ± 0.00	**0.19 ± 0.02**	1
	DinoV2	Climplicit	DA-V2	**0.19 ± 0.02**	*0.32 ± 0.01*	*0.31 ± 0.02*	*0.18 ± 0.05*	4
ST	DinoV2	Climplicit	DA-V2	0.12 ± 0.01	**0.34 ± 0.01**	**0.33 ± 0.01**	**0.21 ± 0.02**	–

Takeaway:
Geo + depth priors boost robustness across traits; single-task models perform best but cost ~4× more.

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- Citizen science images paired with labels from scientific databases (TRY) enable inference of functional traits
- Uncertainty-aware deep learning improves robustness to noisy citizen science images
- Our model generalizes across species and captures intra-species variation
- Our approach outperforms prior work on most traits in a global benchmark

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Thank You

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PlantTraitNet: An Uncertainty-Aware Multimodal Framework for Global-Scale Plant Trait Inference from Citizen Science Data

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Abstract

Global plant maps of plant traits, such as leaf nitrogen or plant height, are essential for understanding ecosystem processes, including the carbon and energy cycles of the Earth system. However, existing trait maps remain limited by the high cost and sparse geographic coverage of field-based measurements. Citizen science initiatives offer a largely untapped resource to overcome these limitations, with over 50 million geotagged plant photographs worldwide capturing valuable visual information on plant morphology and physiology. In this study, we introduce PlantTraitNet, a multi-modal, multi-task uncertainty-aware deep learning framework that predicts four key plant traits (plant height, leaf area, specific leaf area, and nitrogen content) from citizen science photos using weak supervision. By aggregating individual trait predictions across space, we generate global maps of trait distributions. We validate these maps against independent vegetation survey data (sPlotOpen) and benchmark them against leading global trait products. Our results show that PlantTraitNet consistently outperforms existing trait maps across all evaluated traits, demonstrating that citizen science imagery, when integrated with computer vision and geospatial AI, enables not only scalable but also more accurate global trait mapping. This approach offers a powerful new pathway for ecological research and Earth system modeling.

Code — github.com/GeoSense-Freiburg/PlantTraitNet
Datasets —
huggingface.co/datasets/ayushi3536/PlantTraitNet

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Extended version — https://arxiv.org/abs/2511.06943

Introduction

Terrestrial plants, as the largest primary producers on Earth, contribute about 60% to the global net primary productivity (Field et al. 1998) and play a critical role in the carbon and energy cycles of our Earth system (Pan et al. 2011; Schlesinger and Bernhardt 2020). However, understanding how plants influence these cycles is challenging, as the functioning of plants varies profoundly according to their traits. For instance, traits such as canopy height and leaf area control resource acquisition, while leaf tissue properties, such as nitrogen content or dry matter content, are indicators of plant resilience (Díaz et al. 2016). Although these traits are essential for understanding ecosystem processes, the data on such traits is sparse, as their measurement involves costly field surveys and laboratory analysis. Global plant trait databases such as TRY (Kattge et al. 2011) aggregate thousands of trait measurements from numerous studies and regions, providing an invaluable resource for functional biogeography and ecosystem modeling. However, even with these collective efforts, significant gaps persist in the geographic coverage of trait data across biomes, ecosystems, and species, constraining our ability to fully understand and predict global patterns of vegetation function and change (Díaz et al. 2016; Kattge et al. 2020).

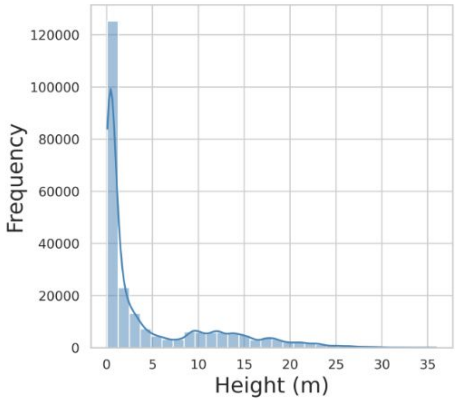
Given the strong link between plant morphology and function, plant photographs in concert with computer vision offer a promising avenue for large-scale estimation of plant traits. Citizen science platforms such as iNaturalist (Su

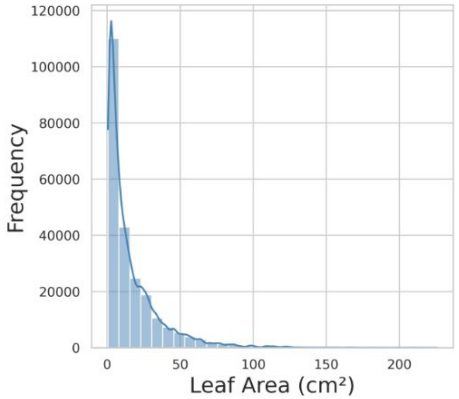
References

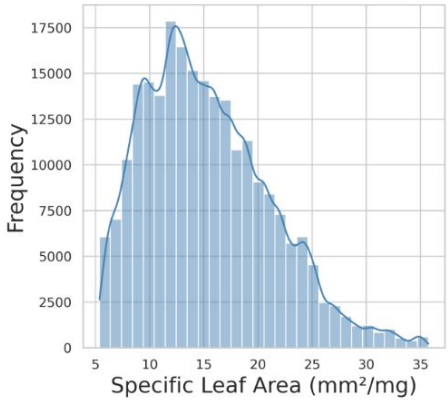
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- Sabatini, F. M. et al. sPlotOpen – An environmentally balanced, open-access, global dataset of vegetation plots. Global Ecology and Biogeography 30, 1740–1764. doi:10.1111/geb. 780 13346 (2021).

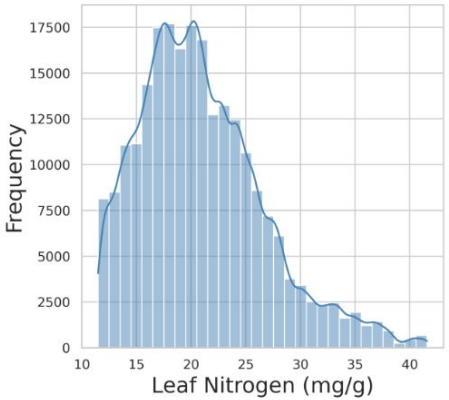
Appendix

Trait Distribution









Reference Data

- Utilized to find the ‘Turning Point’ in the uncertainty-aware framework
- Sparse but reliable

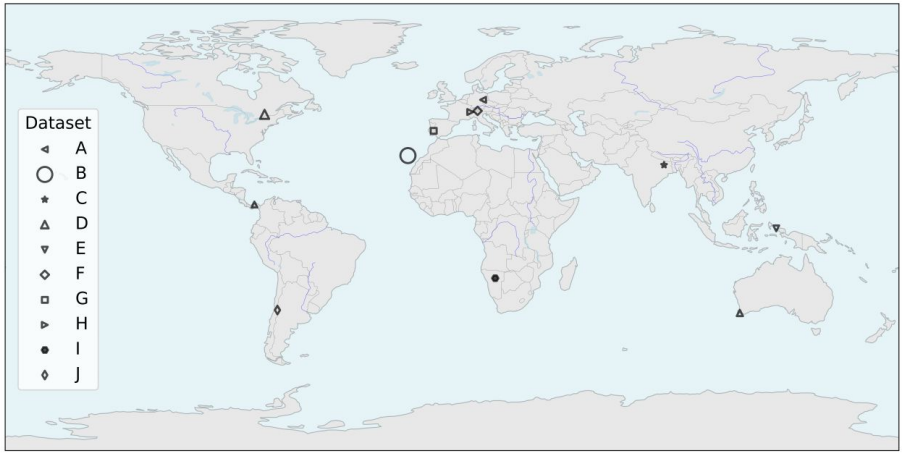


Figure 8: Locations of scientifically curated reference datasets. Marker symbols indicate distinct test data sources from Germany (A), La Palma (B), India (C), Australia (D), Panama (D), Canada (D), Indonesia (E), Switzerland (F), Portugal (G), Switzerland (H), Namibia (I), and Chile (J). Marker sizes are proportional to the corresponding dataset size.

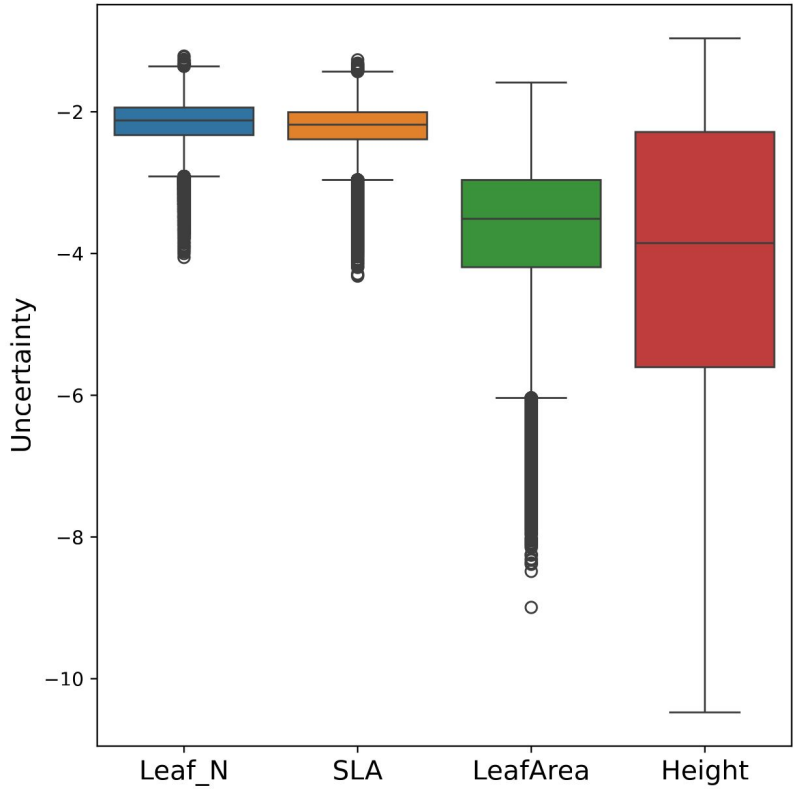
Trait Metadata For Reference Data

Table 3: Metadata of the curated plant reference dataset to aid uncertainty-guided data cleaning.

Trait	Units	#Data points	#Species
Plant height (H)	m	4478	452
Leaf Area (LA)	cm^2	2386	136
Specific Leaf Area (SLA)	mm^2/mg	1796	199
Leaf Nitrogen Content (LN)	mg/g	773	73

Quantitative Assessment of Unceratinty

- More variance for height



Scatter Plot

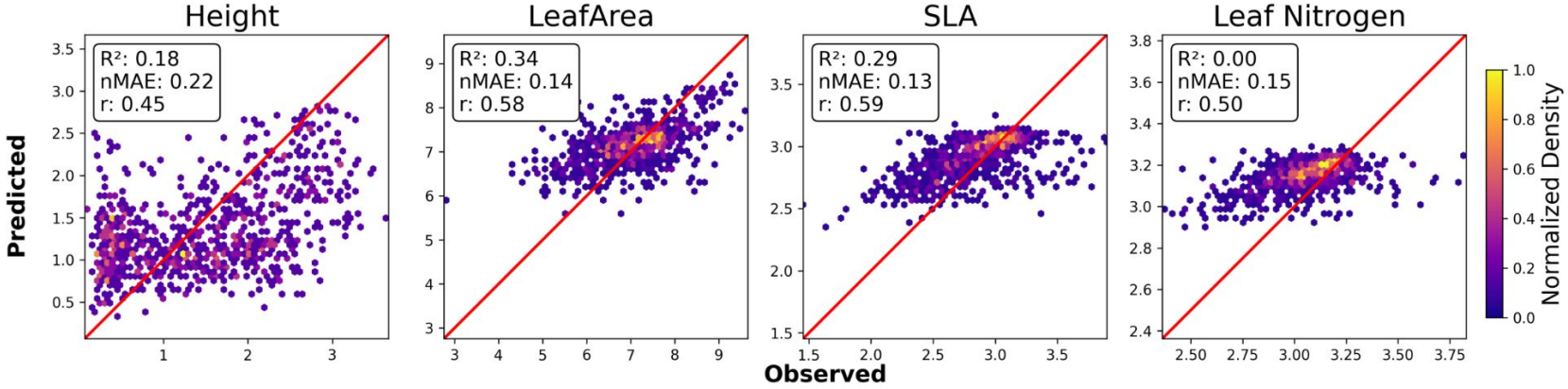
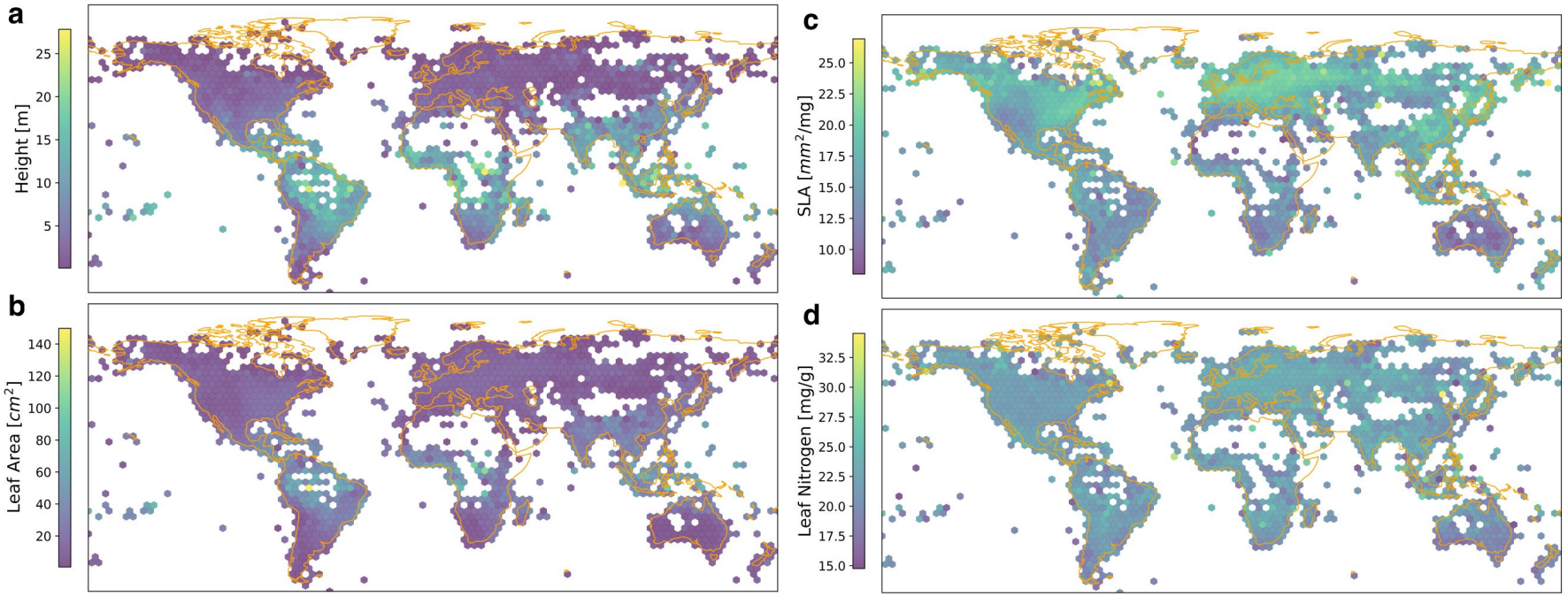
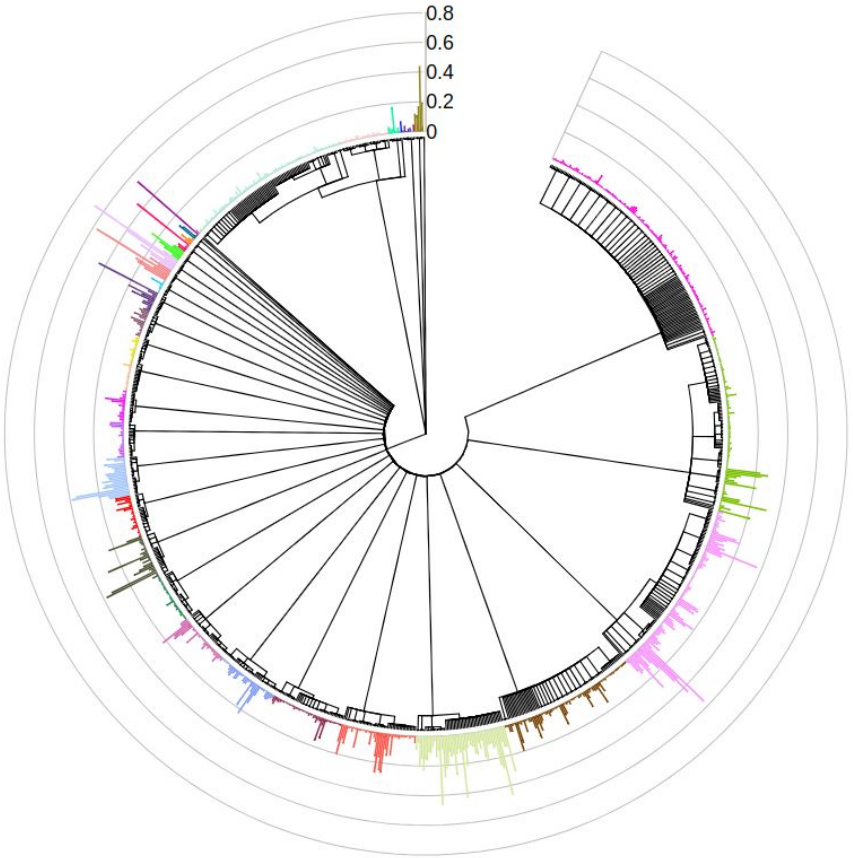


Figure 13: Global trait predictions obtained from PlantTraitNet against globally distributed vegetation survey data (sPlotOpen). Trait predictions of geotagged images are aggregated at a 1° spatial resolution using image geolocations (Predicted). These aggregated predictions are then compared with averaged community-weighted trait values derived from sPlotOpen (Observed).

Global Trait Map Product



Results - Phylogenetic Correlation for Height



Order
- Apiales
- Aquifoliales
- Arecales
- Asparagales
- Asterales
- Boraginales
- Brassicales
- Caryophyllales
- Celastrales
- Cornales
- Cyatheales
- Dipsacales
- Equisetales
- Ericales
- Fabales
- Fagales
- Gentianales
- Geraniales
- Lamiales
- Laurales
- Liliales
- Magnoliales
- Malpighiales
- Malvales
- Myrtales
- Oxalidales
- Pinales
- Piperales
- Poales
- Polypodiales
- Proteales
- Ranunculales
- Rosales
- Sapindales
- Saxifragales
- Solanales

Fagales



Asterales



Rosales



Model

Image Encoder

- **Backbone:** Pretrained **DINOv2 ViT-B/14** for patch-level feature extraction
  (Oquab et al., 2023)

- **Patch embeddings:**
  Z∈RN×C\mathbf{Z} \in \mathbb{R}^{N \times C}Z∈RN×C

- **Pooling:**
  Adaptive average pooling over patch tokens NNN

- **Dimensionality reduction:**
  Compressed to **32 dimensions**
  (parameter-free, preserves coarse spatial structure)

- **Projection head:**
  Flattened pooled features passed through an **MLP**

- **Final image embedding:**
  Ximg∈R768\mathbf{X}_{\text{img}} \in \mathbb{R}^{768}Ximg∈R768

Model

Depth Encoder

- Adds **explicit 3D spatial cues** absent in RGB
- **Encoder:** Frozen **Depth-Anything-V2 (ViT-B)**
- **Output:** Patch embeddings h(I)∈RN×Ch(I) \in \mathbb{R}^{N \times C}h(I)∈RN×C
- **Pooling:** Adaptive average pooling
  → **64-D** representation
- **Projection:** MLP → **768-D depth embedding**
  XdepthX_{\text{depth}}Xdepth

Model

Geo Encoder

Plant traits reflect **local climate adaptation**

Model: Climplicit

Input: Geo-coordinates

Embedding:
Xgeo∈R1024X_{\text{geo}} \in \mathbb{R}^{1024}Xgeo∈R1024

Projection: Linear layer → **256-D**

Model

Multimodal Fusion

- **Concatenation:**
  [Ximg ｜ Xdepth ｜ Proj(Xgeo)][X_{\text{img}} \; | \; X_{\text{depth}} \; | \; \text{Proj}(X_{\text{geo}})][Ximg | Xdepth | Proj(Xgeo)]
- **Fusion projection:** Linear → **1024-D**

- **Backbone:** 8-block residual network
  (hidden dim = 2×2 \times2× embedding size)

- **Output:** Shared multimodal representation